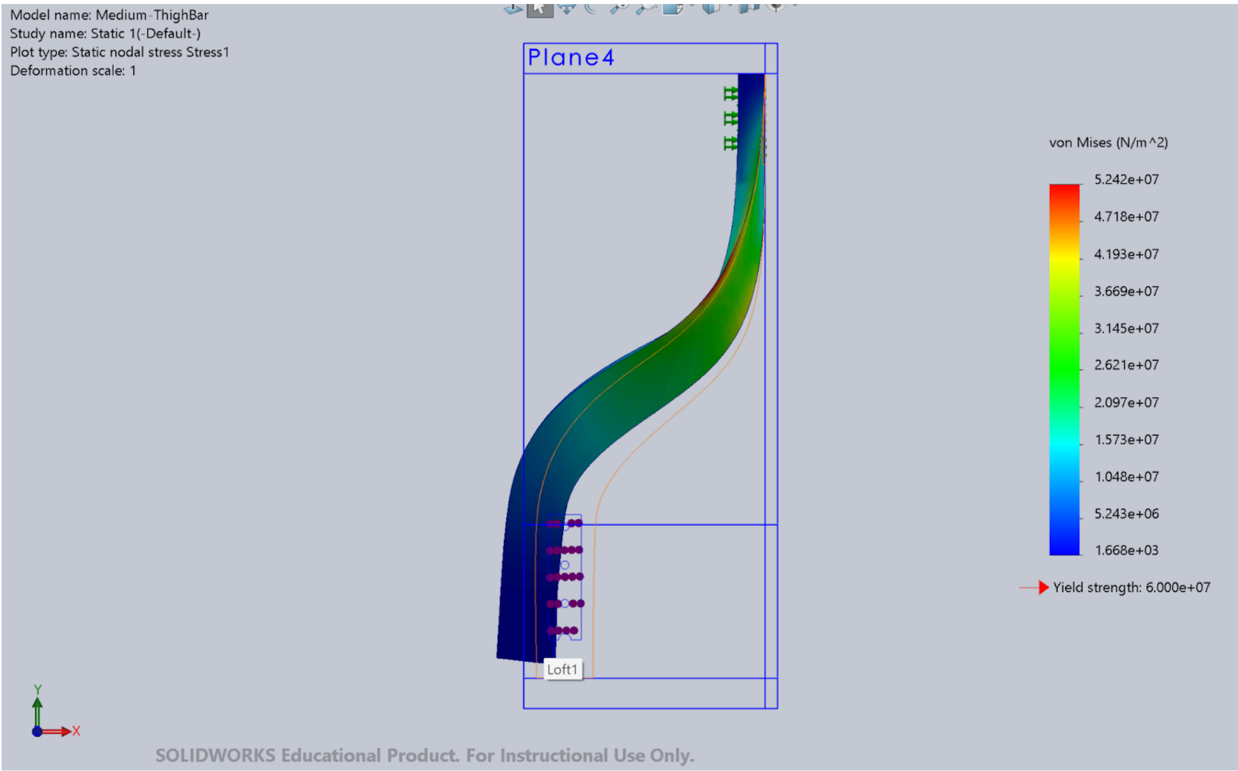
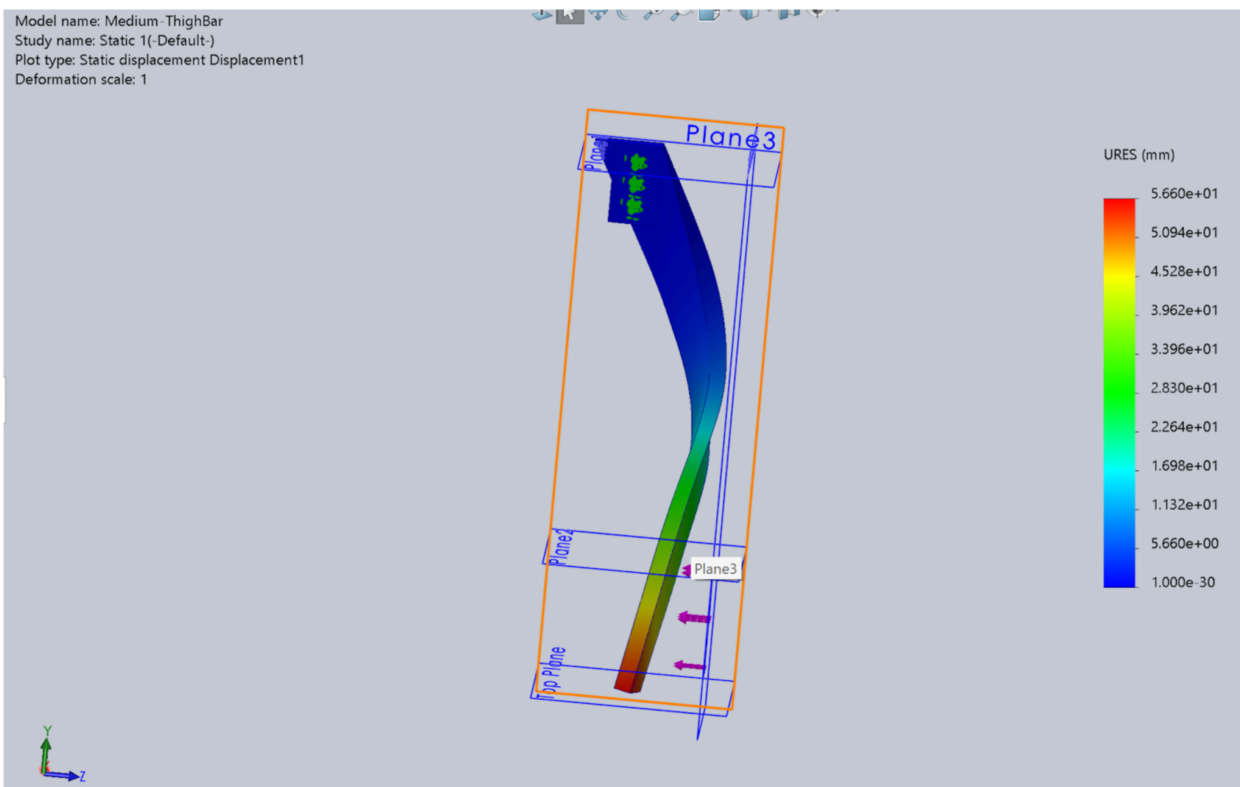
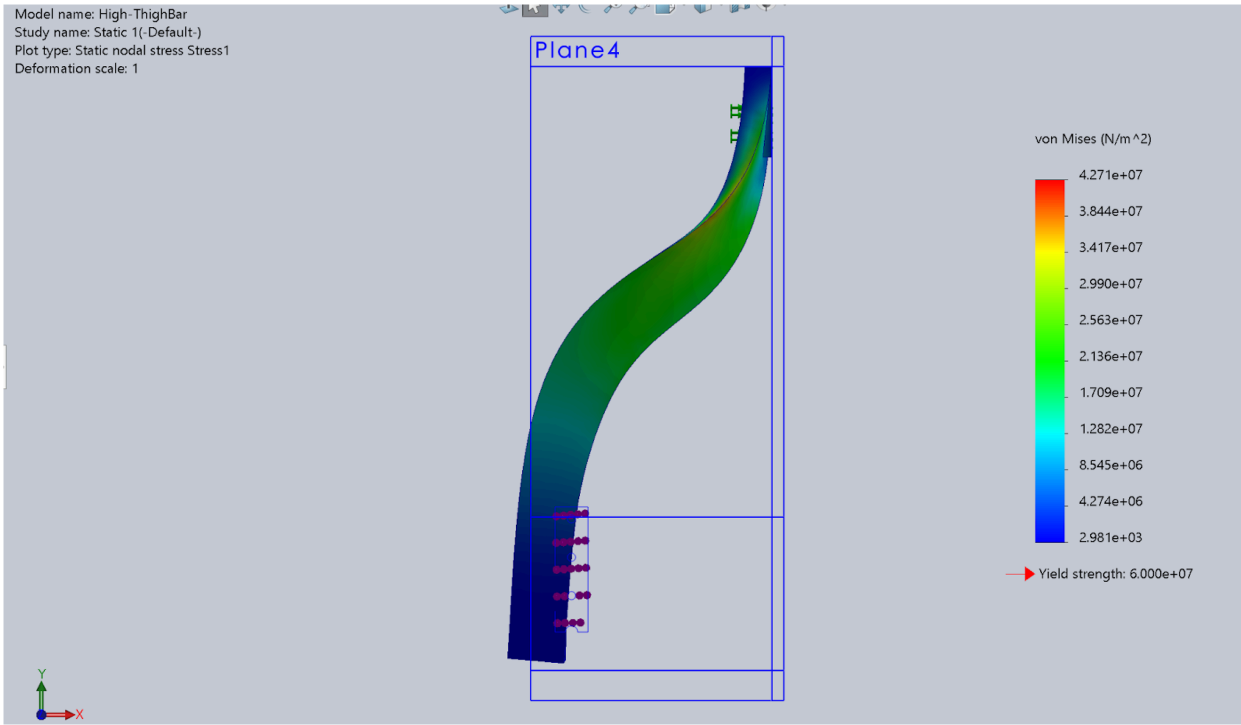
Medium-ThighBar

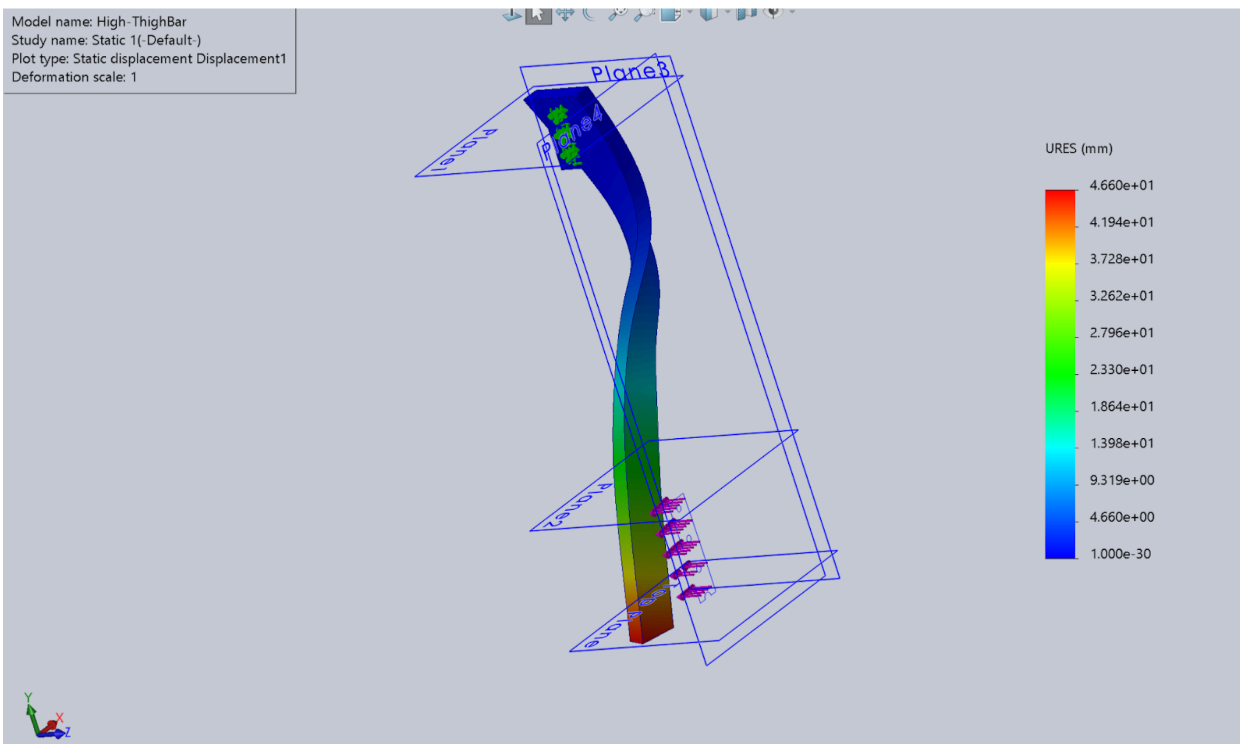




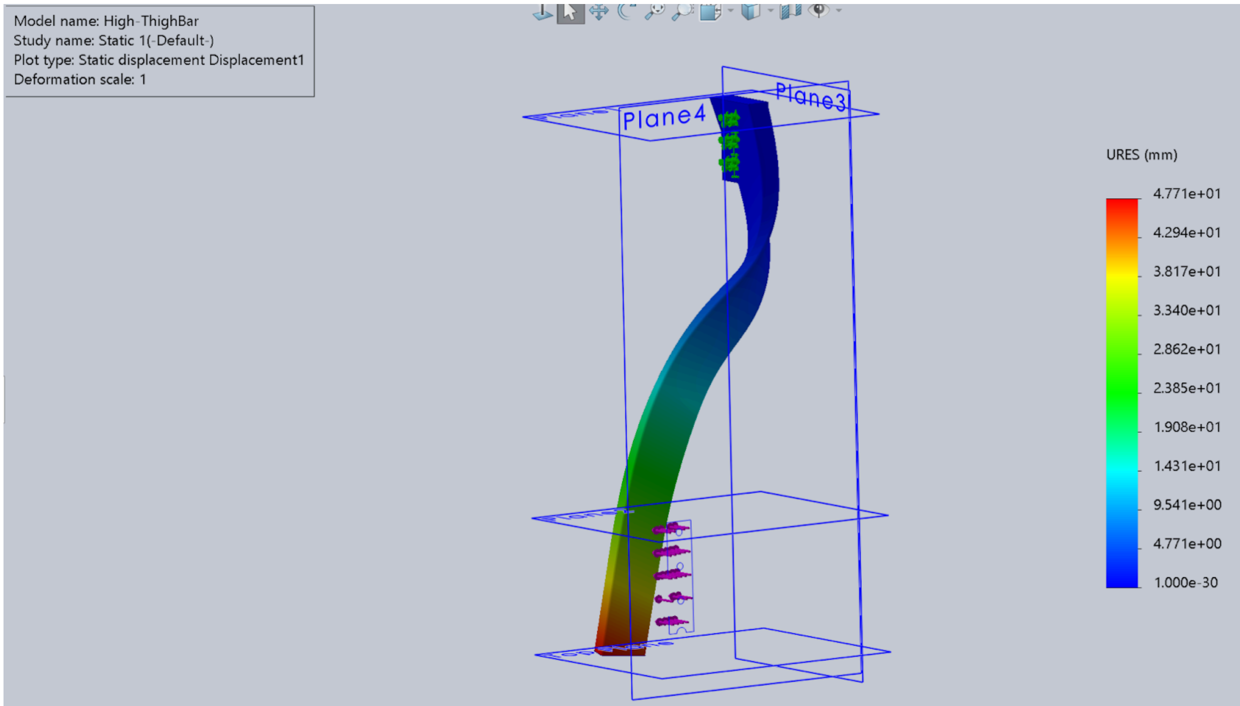
High-ThighBar



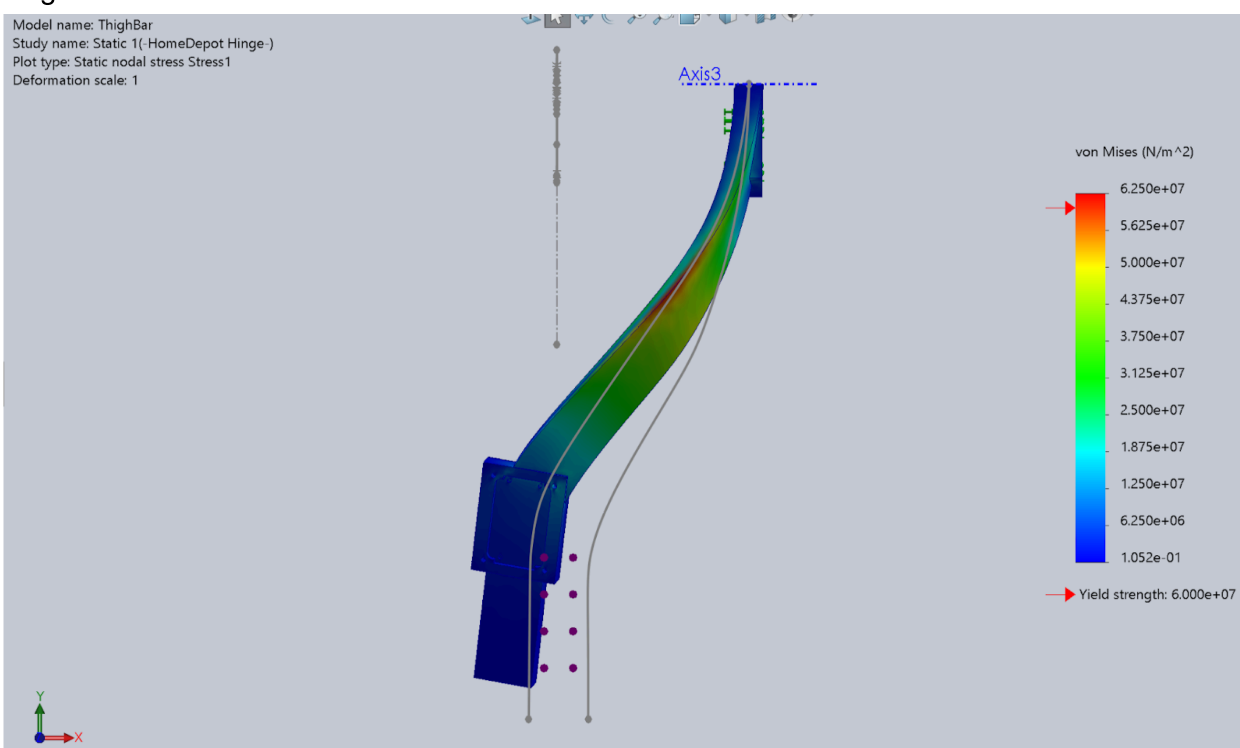
Displacement (Without fillets):



With fillets


Model name: High-ThighBar
Study name: Static 1(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 1
Plane4
Plane3
URES (mm)
4.771e+01
4.294e+01
3.817e+01
3.340e+01
2.862e+01
2.385e+01
1.908e+01
1.431e+01
9.541e+00
4.771e+00
1.000e-30

ThighBar


Model name: ThighBar
Study name: Static 1(-HomeDepot Hinge-)
Plot type: Static nodal stress Stress1
Deformation scale: 1
Axis3
von Mises (N/m^2)
6.250e+07
5.625e+07
5.000e+07
4.375e+07
3.750e+07
3.125e+07
2.500e+07
1.875e+07
1.250e+07
6.250e+06
1.052e-01
Yield strength: 6.000e+07
Y
X

Why does setting the curvature higher result in smaller von mises stresses?

- Maximum Principal Stress Theory (because PLA is brittle)

  The principal stresses (σ_1, σ_2) are the maximum and minimum normal stresses acting on a material element, calculated using:

  $$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

  - **σ_1 (Maximum Principal Stress):** Use the plus sign (+).
  - **σ_2 (Minimum Principal Stress):** Use the minus sign (-). StudySmarter UK

- For the medium bar, the force produces both bending and torsional stress, contributing to a larger combined stress
- For the high bar, the curvature decreases torsional stress, which contributes on a greater scale than bending stress, thus minimizing the overall combined stress
  - long lower section acts like a cantilever beam, resulting in less torsional stress on the curve (less leverage)